LOWESS/Spline Smoothing Data Application and Analysis Project

Dataset: Bike Sharing Dataset

Goal: Apply LOWESS and Spline smoothing to this dataset, and analyze their impact on understanding the raw data, and compare their outcomes to each other.

1. Setup

In [16]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from statsmodels.nonparametric.smoothers_lowess import lowess
from scipy.interpolate import UnivariateSpline
```

2. Load Dataset

In [17]:
```
plt.rcParams['figure.figsize'] = (8,5)

df = pd.read_csv('hour.csv')

df = df[['hr','cnt']]
df.dropna(inplace=True)
df
```

Out[17]:

	hr	cnt
0	0	16
1	1	40
2	2	32
3	3	13
4	4	1
...	...	...
17374	19	119
17375	20	89
17376	21	90
17377	22	61
17378	23	49

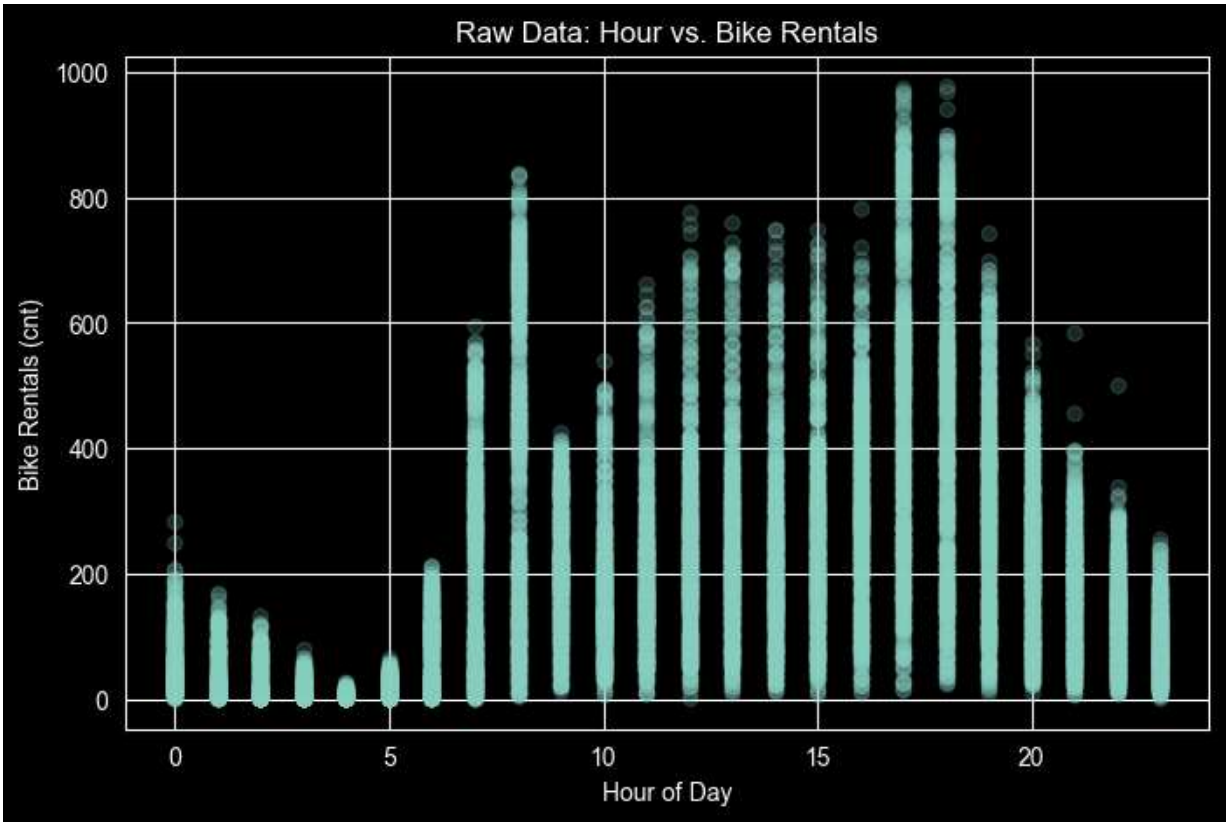
17379 rows × 2 columns

3. Raw Data Visualization

Visualizing Hour vs Number of Bike Rentals

In [18]:

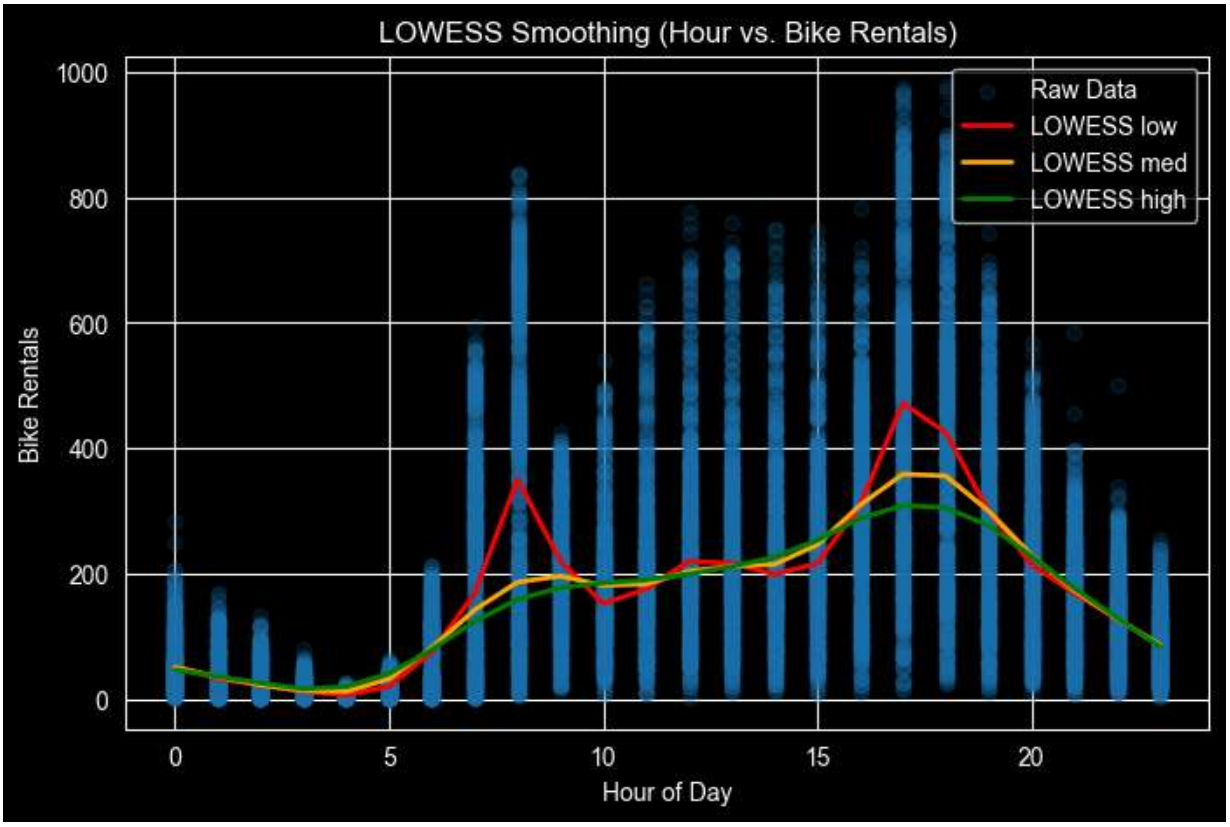
```
plt.scatter(df['hr'],df['cnt'], alpha=0.2)
plt.xlabel('Hour of Day')
plt.ylabel('Bike Rentals (cnt)')
plt.title('Raw Data: Hour vs. Bike Rentals')
plt.show()
```



4. LOWESS Smoothing

Applying LOWESS smoothing to the Hour vs. Bike Rentals data.

In [19]:
```
#showcasing frac factor
lowess_hr_low = lowess(df['cnt'], df['hr'], frac=0.05)
lowess_hr_med = lowess(df['cnt'], df['hr'], frac=0.15)
lowess_hr_high = lowess(df['cnt'], df['hr'], frac=0.25)

#plot scatter
plt.scatter(df['hr'],df['cnt'], alpha=0.15, color='tab:blue', label='Raw Data')

#plot lowess output
plt.plot(lowess_hr_low[:,0],lowess_hr_low[:,1], color='red', linewidth=2, label='LO
plt.plot(lowess_hr_med[:,0],lowess_hr_med[:,1], color='orange', linewidth=2, label=
plt.plot(lowess_hr_high[:,0],lowess_hr_high[:,1], color='green', linewidth=2, label

plt.xlabel('Hour of Day')
plt.ylabel('Bike Rentals')
plt.title('LOWESS Smoothing (Hour vs. Bike Rentals)')
plt.legend()
plt.show()
```

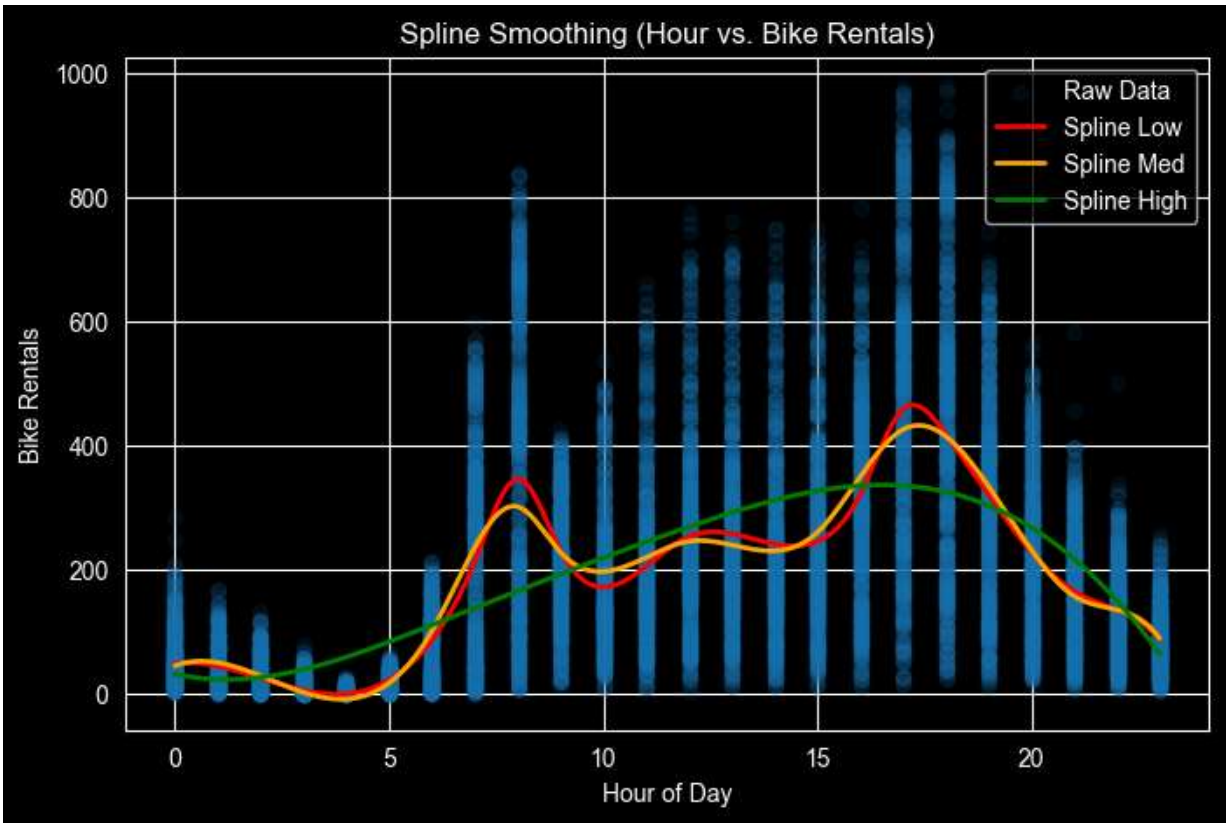


5. Spline Smoothing

Spline Smoothing for Hour vs. Num Bike Rentals

In [24]:
```
# aggregate data for spline
df_grouped = df.groupby('hr')['cnt'].mean().reset_index()

x = df_grouped['hr']
y = df_grouped['cnt']

# fit spline
from scipy.interpolate import UnivariateSpline
spline_low = UnivariateSpline(x, y, s=1000)
spline_med = UnivariateSpline(x, y, s=10000)
spline_high = UnivariateSpline(x, y, s=100000)

# generate curve
x_range = np.linspace(x.min(), x.max(), 1000)
y_spline_low = spline_low(x_range)
y_spline_med = spline_med(x_range)
y_spline_high = spline_high(x_range)

# plot
plt.scatter(df['hr'], df['cnt'], alpha=0.1, color='tab:blue', label='Raw Data')
plt.plot(x_range, y_spline_low, color='red', linewidth=2, label='Spline Low')
plt.plot(x_range, y_spline_med, color='orange', linewidth=2, label='Spline Med')
plt.plot(x_range, y_spline_high, color='green', linewidth=2, label='Spline High')

plt.xlabel('Hour of Day')
plt.ylabel('Bike Rentals')
```

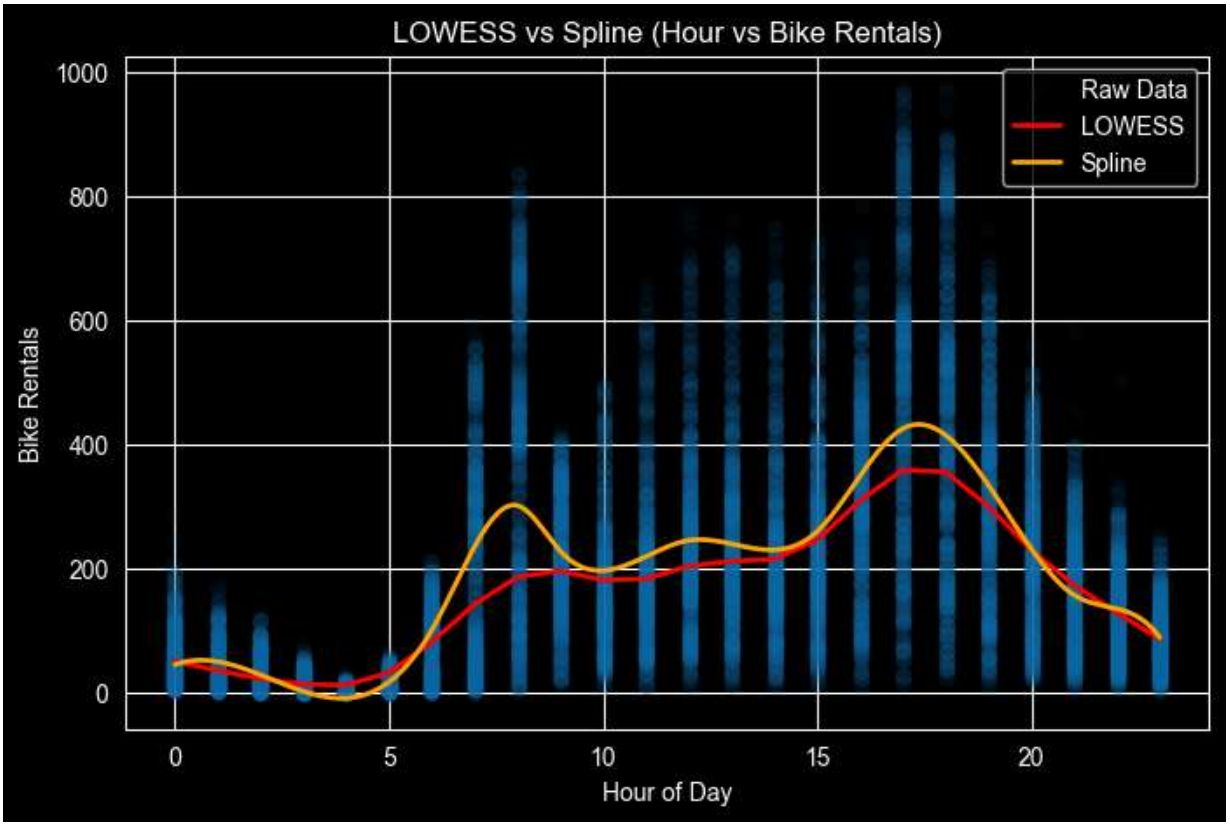
```
plt.title('Spline Smoothing (Hour vs. Bike Rentals)')
plt.legend()
plt.show()
```



6. Direct Comparison

Spline vs. LOWESS

In [21]:
```
plt.scatter(df['hr'], df['cnt'], alpha=0.05, color='tab:blue', label='Raw Data')

plt.plot(lowess_hr_med[:, 0], lowess_hr_med[:, 1],
         color='red', linewidth=2, label='LOWESS')

plt.plot(x_range, y_spline_med,
         color='orange', linewidth=2, label='Spline')

plt.xlabel('Hour of Day')
plt.ylabel('Bike Rentals')
plt.title('LOWESS vs Spline (Hour vs Bike Rentals)')
plt.legend()
plt.show()
```


LOWESS vs Spline (Hour vs Bike Rentals)
Raw Data
LOWESS
Spline
Bike Rentals
1000
800
600
400
200
0
0
5
10
15
20
Hour of Day